

DATA PRIVACY **AND SECURITY**

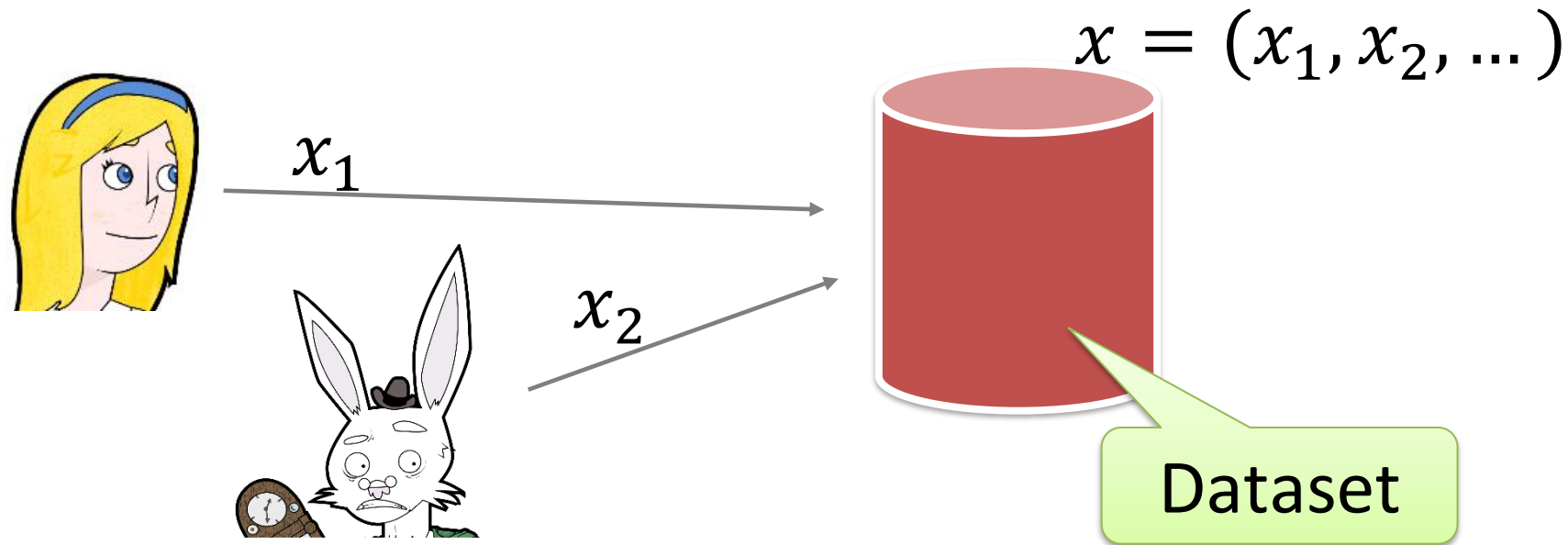
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Chapter 4: Differential Privacy

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Data Exploitation



- Availability of **lots of data**
 - Social networks, financial data, medical records...
- All these data are an asset
 - We would like to **exploit them**

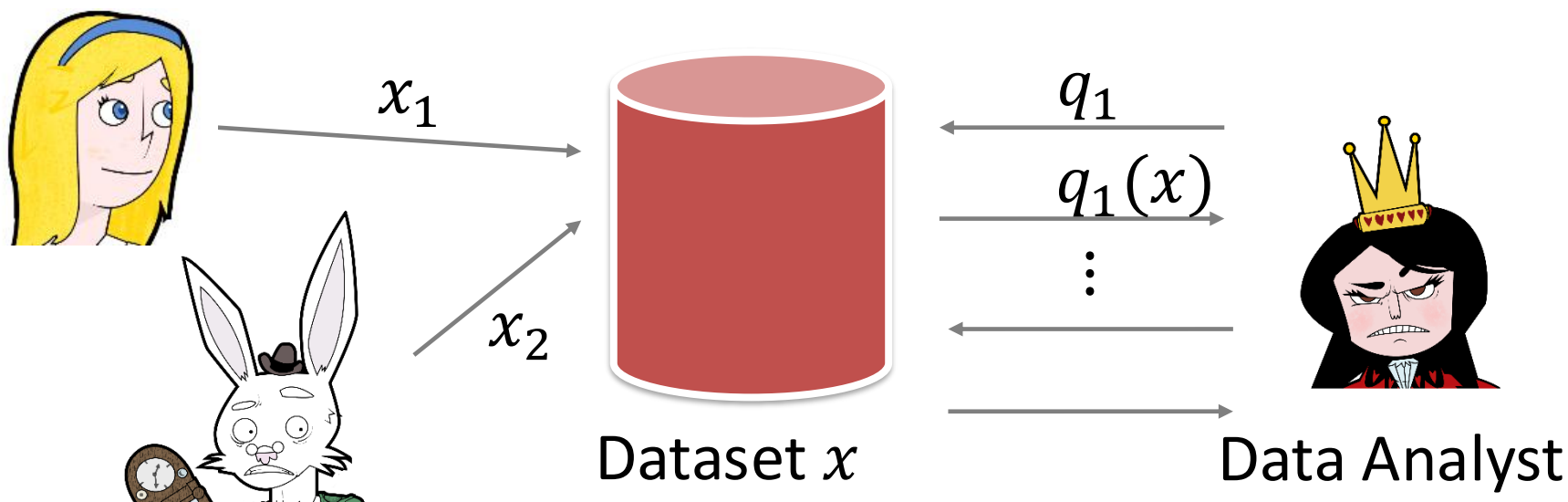


Applications

- Finding **statistical correlations**
 - Genotype/phenotype association
 - Correlating medical outcomes with risk factors
- Publishing **aggregate** statistics
- Noticing **events/outliers**
 - Intrusion detection
- Data mining/**learning**
 - Update **strategies** based on customers data



Data Analysis and Privacy



- How to define **privacy**?
 - Intuitively we want that published statistics do not undermine privacy of the individuals
 - After all statistics are just **aggregated data** about the overall population



The Statistics Masquerade

- Differential attack
 - How many people in the room XYZ last night?
 - How many people, **other than the speaker**, XYZ last night?
- Needle in a haystack
 - Determine presence of an individual genomic data in GWAS case group based on **aggregate stats**
- The big bang attack
 - Reconstruct sensitive attributes given statistics from **multiple overlapping datasets**



NYC Taxicab Data

- 2014: NYC Taxi & Limo Commission sharing visualization on **taxi usage statistics** on twitter
 - Chris Whong filed a FOIL request and released the dataset **publicly online**
 - 19 GB with all taxi fares and statistics in 2013
- Attempt to **anonymize** the data
6B111958A39B24140C973B262EA9FEA5,
D3B035A03C8A34DA17488129DA581EE7, ...
 - Someone discovered those were the MD5 hash of the driver's **medallion** and **license number**



The Netflix Prize

- 2006-2009: 1M USD for improving the **recommendation engine**
- **Anonymized** dataset including movie id, user id, rating and date
- The dataset was **de-anonymized** by combining it with the **public** IMDB dataset
 - Matching users that gave **similar** preferences
 - A class action **lawsuit** was filed against Netflix

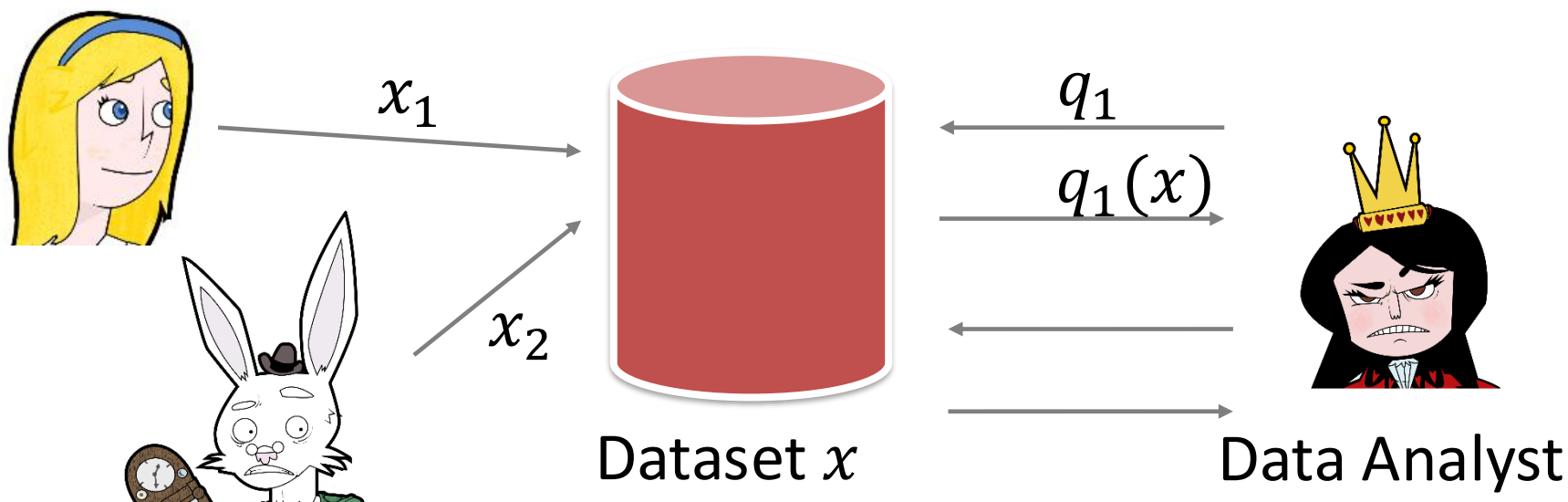


Lessons to be Learned

- Privacy is a **concern** when publishing datasets
- Wait: This does not apply to me!
 - Don't make the **entire** dataset available
 - Only publish **aggregate** statistics
- Even if only **data aggregations** are published privacy **can be broken**
- Overly accurate estimates of too many statistics is **blatantly non-private**



Privacy-Preserving Data Analysis?



- Can't learn **anything new** about Alice?
 - Reminiscent of semantic security for encryption
- Learn the same thing if Alice is **replaced** by a **random member** of the population

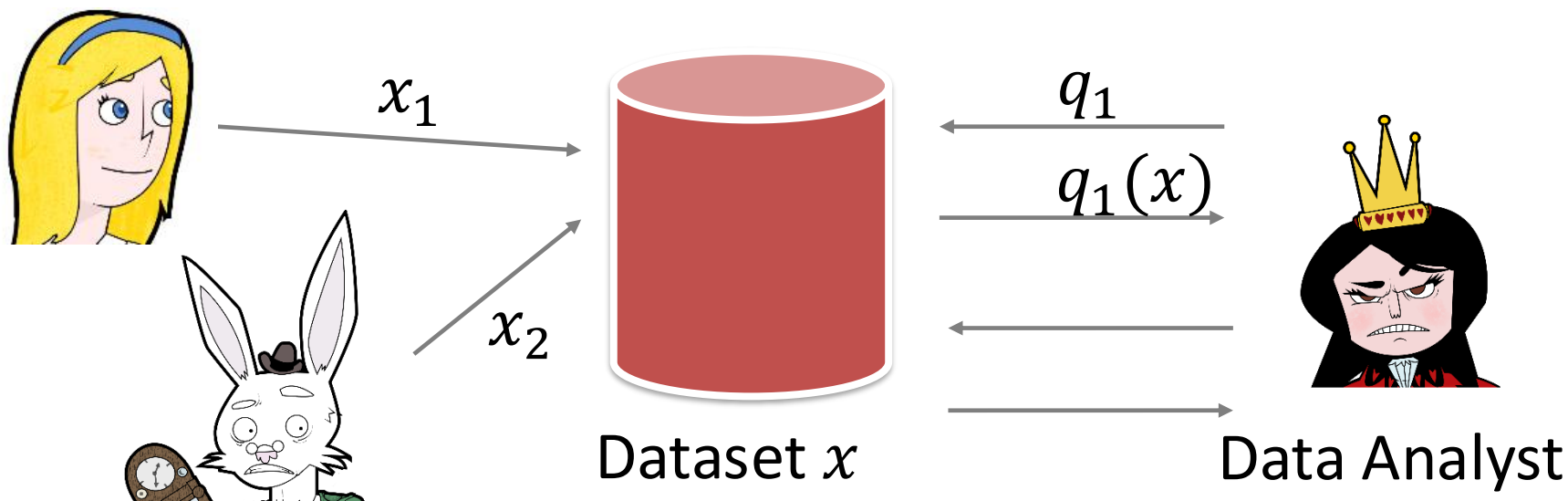


Differential Privacy

- Outcome of analysis is roughly **equally likely**
 - **Independent** of whether any individual joins, or refrains from joining, the dataset
 - Alice goes away, Bob joins, Alice replaced by Bob
 - Small perturbations **do not matter**
- Note that instead if we **completely change** the dataset, we get **completely different** answers!
- Adopted in **real-world** applications by Apple, Google and Microsoft



More Formally...



Definition: Mechanism $\mathbf{M}$: $\mathcal{X}^n \times \mathcal{Q} \rightarrow \mathcal{Y}$ gives ε -**differential privacy** if for all pairs of **adjacent** datasets $x, x' \in \mathcal{X}^n$, and for all queries $q \in \mathcal{Q}$:

$$\forall y \in \mathcal{Y}, \mathbb{P}[\mathbf{M}(x, q) = y] \leq e^\varepsilon \cdot \mathbb{P}[\mathbf{M}(x', q) = y]$$


Notes on the Definition

- All that an adversary learns about **you**, it could have learned from the **rest of the dataset**
 - Even if you don't participate
 - This **doesn't mean** nothing about you is leaked
 - Can't use DP to take actions on **specific** individuals
- **Worst-case** guarantee
 - For all datasets, against **unbounded** adversaries
- Probability over the **randomness** of the algorithm, not over the choice of the dataset



Counting Queries

- Simply a **predicate on rows** $q: \mathcal{X} \rightarrow \{0,1\}$
 - Can be extended to datasets $\mathcal{X}^n$ by counting the fraction of people satisfying the predicate, i.e.

$$q(x) = \frac{1}{n} \sum_{i=1}^n q(x_i)$$

- **Point functions**: $\mathcal{Q}_{\text{pt}}(\mathcal{X}) = \{q_y\}_{y \in \mathcal{X}}$ s.t. $q_y(w) = 1$ iff $w = y$
 - Answering all queries in $\mathcal{Q}_{\text{pt}}(\mathcal{X})$ amounts to computing the **histogram** of the dataset



Counting Queries

- **Threshold functions:** $\mathcal{Q}_{\text{thr}}(\mathcal{X}) = \{q_y\}_{y \in \mathcal{X}}$ s.t. $q_y(w) = 1$ iff $w \leq y$ (with $\mathcal{X}$ totally ordered)
 - Answering all queries in $\mathcal{Q}_{\text{thr}}(\mathcal{X})$ amounts to the **cumulative distribution function** of the dataset
- **Attribute means:** $\mathcal{Q}_{\text{means}}(d) = \{q_j\}_{j \in [d]}$ s.t. $q_j(w) = w_j$, where $w \in \mathcal{X} = \{0,1\}^d$
 - Answering all queries in $\mathcal{Q}_{\text{means}}(d)$ amounts to computing the **fraction** of the dataset possessing each of the d attributes (1-way marginal statistics)



Counting Queries

- **Conjunctions:** $Q_{\text{conj}}^t(d)$ with all conjunctions of $t \in [0, d]$ literals on $\mathcal{X} = \{0,1\}^d$
 - E.g., $Q_{\text{conj}}^2(5)$ contains $q(w) = w_2 \wedge \neg w_4$ (what fraction of individual in the dataset have lung cancer and are non-smokers?)
 - These are called ***t*-way marginal**
 - Answering all queries in $Q_{\text{conj}}^t(d)$ amounts to computing the ***t*-way contingency table**



Postprocessing

- **Theorem:** If $\mathbf{M}: \mathcal{X}^n \times \mathcal{Q} \rightarrow \mathcal{Y}$ is ε -DP, and $\Psi: \mathcal{Y} \rightarrow \mathcal{Z}$ is any **randomized function**, then $\Psi \circ \mathbf{M}: \mathcal{X}^n \times \mathcal{Q} \rightarrow \mathcal{Z}$ is ε -DP
- Let Ψ be a **distribution** on **deterministic** $\psi: \mathcal{Y} \rightarrow \mathcal{Z}$.
For any $z \in \mathcal{Z}$:

$$\begin{aligned} & \mathbb{P}[(\Psi \circ \mathbf{M})(x) = z] \\ &= \mathbb{E}_{\psi \leftarrow \Psi} [\mathbb{P}[\mathbf{M}(x) = \psi^{-1}(z)]] \\ &\leq \mathbb{E}_{\psi \leftarrow \Psi} [e^\varepsilon \cdot \mathbb{P}[\mathbf{M}(x') = \psi^{-1}(z)]] \\ &= e^\varepsilon \cdot \mathbb{P}[(\Psi \circ \mathbf{M})(x') = z] \end{aligned}$$



Group Privacy

- **Theorem:** If $\mathbf{M}$ is ε -DP, then for all pairs of datasets $x, x' \in \mathcal{X}^n$, $\mathbf{M}(x)$ and $\mathbf{M}(x')$ are $k\varepsilon$ -DP for $k = d(x, x')$
 - Here, $d(x, x')$ is the number of rows that **need to be changed** to go from x to x'
 - Let x_{i+1} be obtained from x_i by changing **one row**

$$\begin{aligned}\mathbb{P}[\mathbf{M}(x_0) = y] &\leq e^\varepsilon \cdot \mathbb{P}[\mathbf{M}(x_1) = y] \\ &\leq e^\varepsilon \cdot (e^\varepsilon \cdot \mathbb{P}[\mathbf{M}(x_2) = y]) \\ &\vdots \\ &\leq e^{k\varepsilon} \cdot \mathbb{P}[\mathbf{M}(x_k) = y]\end{aligned}$$



Basic Composition

- **Theorem:** If $\mathbf{M}_1, \dots, \mathbf{M}_k$ are ε -DP, then $\mathbf{M}$ s.t. $\mathbf{M}(x) = (\mathbf{M}_1(x), \dots, \mathbf{M}_k(x))$ is $k\varepsilon$ -DP
- Fix $x \sim x'$. For $y \in \mathcal{Y}$, define

$$\Lambda_{\mathbf{M}(x) || \mathbf{M}(x')}(y) = \ln \left(\frac{\mathbb{P}[\mathbf{M}(x) = y]}{\mathbb{P}[\mathbf{M}(x') = y]} \right)$$

- When $\Lambda_{\mathbf{M}(x) || \mathbf{M}(x')}(y) > 0$, the outcome y is "**evidence**" that the dataset is x rather than x'
- Thus, ε -DP means that for all $x \sim x'$, and for all y ,
 $|\Lambda_{\mathbf{M}(x) || \mathbf{M}(x')}(y)| \leq \varepsilon$



Basic Composition

- In our case:

$$\begin{aligned} & \Lambda_{\mathbf{M}(x) || \mathbf{M}(x')}(y) \\ &= \ln \left(\frac{\mathbb{P}[\mathbf{M}_1(x) = y_1 \wedge \dots \wedge \mathbf{M}_k(x) = y_k]}{\mathbb{P}[\mathbf{M}_1(x') = y_1 \wedge \dots \wedge \mathbf{M}_k(x') = y_k]} \right) \\ &= \ln \left(\frac{\prod_{i=1}^k \mathbb{P}[\mathbf{M}_i(x) = y_i]}{\prod_{i=1}^k \mathbb{P}[\mathbf{M}_i(x') = y_i]} \right) \\ &= \sum_{i=1}^k \Lambda_{\mathbf{M}_i(x) || \mathbf{M}_i(x')}(y_i) \\ &\Rightarrow |\Lambda_{\mathbf{M}(x) || \mathbf{M}(x')}(y)| \leq \sum_{i=1}^k |\Lambda_{\mathbf{M}_i(x) || \mathbf{M}_i(x')}(y_i)| \leq k\varepsilon \end{aligned}$$



Summary of Properties

- Immune to **auxiliary information**
 - Current and future side information
- Automatically yields **group privacy**
 - Privacy loss $k\varepsilon$ for groups of size k
- Composition
 - Can bound cumulative privacy loss over **multiple analysis** (the epsilons add up)
 - Can combine a few differentially private mechanisms to solve **complex analytical tasks**



Did You XYZ Last Night?

- Flip a coin
 - If heads, **flip again** and return YES if heads, and else return NO
 - If tails, **answer honestly**
- $\frac{\mathbb{P}[\text{YES}|\text{Truth}=\text{YES}]}{\mathbb{P}[\text{YES}|\text{Truth}=\text{NO}]} = \frac{1/2 + 1/2 \cdot (1/2 + 0)}{0 + 1/2 \cdot (1/2)} = 3$
- $\frac{\mathbb{P}[\text{NO}|\text{Truth}=\text{NO}]}{\mathbb{P}[\text{NO}|\text{Truth}=\text{YES}]} = 3$
- Gives ϵ -DP for $\epsilon = \ln 3 \approx 1.098$
- Expected #YES: $1/4 (1 - p) + 3/4 p$

p = fraction of people that XYZ
 $= 2(\text{\#YES}) - 1/4$



Randomized Response: Privacy

- Let $q: \mathcal{X} \rightarrow \{0,1\}$ be a **counting query**
- For each row x_i , let $\mathbf{M}(x_i) = q(x_i)$ w.p. $1/2 + \varepsilon$ and $\mathbf{M}(x_i) = \overline{q(x_i)}$ w.p. $1/2 - \varepsilon$
- Consider $\mathbf{M}(x) = \mathbf{M}(x_1, \dots, x_n) = (y_1, \dots, y_n)$
 - Assume $x \sim x'$ **differ only** in the j -th row

$$\begin{aligned} \frac{\mathbb{P}[\mathbf{M}(x) = y]}{\mathbb{P}[\mathbf{M}(x') = y]} &= \frac{\prod_i \mathbb{P}[\mathbf{M}(x_i) = y_i]}{\prod_i \mathbb{P}[\mathbf{M}(x'_i) = y_i]} \\ &= \frac{\mathbb{P}[\mathbf{M}(x_j) = y_j]}{\mathbb{P}[\mathbf{M}(x'_j) = y_j]} \leq \frac{1/2 + \varepsilon}{1/2 - \varepsilon} \end{aligned}$$



Randomized Response: Accuracy

- The latter is $\leq e^{O(\varepsilon)}$ when, say, $\varepsilon \leq 1/4$ and thus the mechanism **M** has $O(\varepsilon)$ -DP
- As for **accuracy**, note that
 - $\mathbb{E}[y_i] = (1/2 + \varepsilon) \cdot q(x_i) + (1/2 - \varepsilon) \cdot (1 - q(x_i)) = 2\varepsilon \cdot q(x_i) + 1/2$
 - Thus, $q(x_i) = 1/(2\varepsilon) \cdot \mathbb{E}[(y_i - 1/2 + \varepsilon)]$
- This suggests the following **estimator**

$$\tilde{y} = \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{2\varepsilon} \cdot (y_i - 1/2 + \varepsilon) \right]$$

$$\mathbb{E}[\tilde{y}] = q(x)$$



Randomized Response: Accuracy

- Next, we analyze the **variance**

$$\begin{aligned} - V[\tilde{y}] &= V\left[\frac{1}{n} \sum_{i=1}^n \left[\frac{1}{2\varepsilon} \cdot (y_i - 1/2 + \varepsilon)\right]\right] = \frac{1}{4\varepsilon^2 n^2} \cdot \sum_{i=1}^n V[y_i] \leq \frac{1}{4\varepsilon^2 n} \\ &= \frac{1}{16\varepsilon^2 n} \end{aligned}$$

- Finally, by **Chebyshev's** inequality

$$|\tilde{y} - y| \leq O\left(\frac{1}{\sqrt{n} \cdot \varepsilon}\right)$$

- As $n \rightarrow \infty$, we get an **increasingly accurate** estimate of the **result**



Differential Privacy by Adding Noise

- Let q be a **counting query**
- Answer with $\mathbf{M}(x) = q(x) + \mathbf{noise}$
 - But according to which distribution?
- Note that if $x \sim x'$, then $|q(x) - q(x')| \leq 1/n$
- At every y , the density of output distribution should be same for x, x' **up to a factor** e^ϵ
 - Density of $\mathbf{M}(x)$ (resp. $\mathbf{M}(x')$) at y is that of the noise at $z = y - q(x)$ (resp. $z = y - q(x')$)
 - Again, $|z - z'| \leq 1/n$



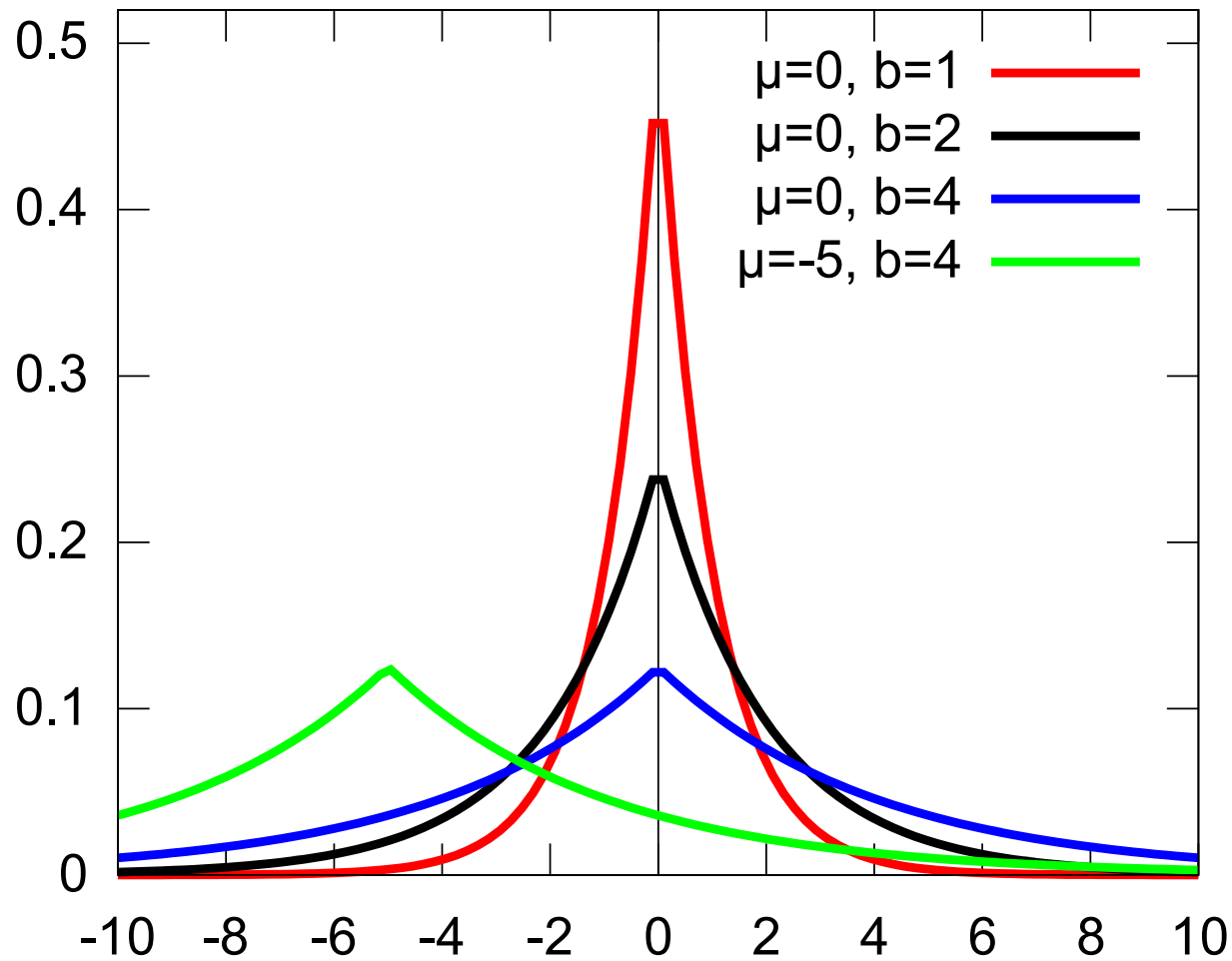
Laplace Mechanism: Privacy

- Let $L(\mu, \sigma)$ at z be $1/(2\sigma) \cdot e^{-|z-\mu|/\sigma}$
- If we set $\mu = 0, \sigma = 1/\varepsilon n$, we have:

$$\begin{aligned} & \frac{\mathbb{P}[\mathbf{M}(x) = y]}{\mathbb{P}[\mathbf{M}(x') = y]} \\ &= e^{\frac{|y-q(x')| - |y-q(x)|}{\sigma}} \\ &\leq e^{\frac{|q(x)-q(x')|}{\sigma}} \leq e^{\frac{1}{n\sigma}} = e^\varepsilon \end{aligned}$$



Laplace Distribution



Laplace Mechanism: Accuracy

- Note that $L(0, \sigma)$ has mean 0 and standard deviation $\sqrt{2}\sigma$, and **exponentially vanishing tails**: $\mathbb{P}[|L(0, \sigma)| > \sigma t] \leq e^{-t}$
- Hence, for any $0 < \beta \leq 1$:

$$\mathbb{P}[|q(x) - y| > \ln(1/\beta) \cdot 1/(\varepsilon n)] \leq \beta$$

- With **high probability** we get error $O(1/(\varepsilon n))$
 - Compare this with accuracy $O(1/\varepsilon\sqrt{n})$ of randomized responses



Laplace Mechanism: Multivariate Case

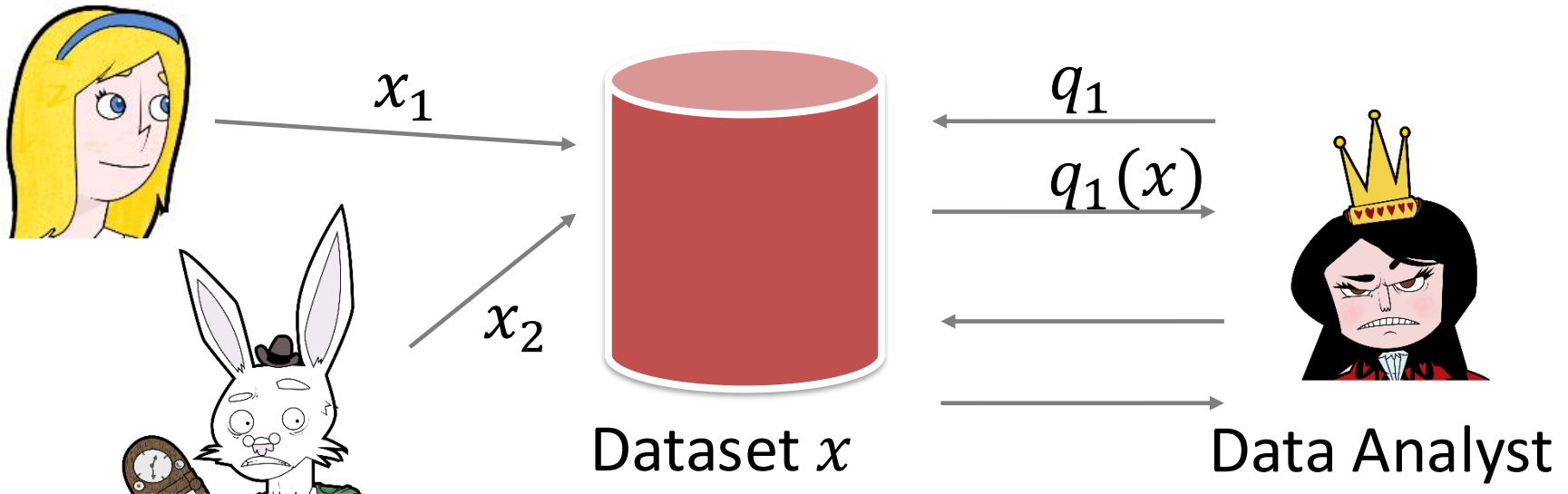
- **Not specific** to counting queries
 - All we used is that $|q(x) - q(x')| \leq 1/n$ for $x \sim x'$
- For arbitrary $q: \mathcal{X}^n \rightarrow \mathbb{R}^d$ scale the noise to **global ℓ_1 -sensitivity**

$$\Delta_1 = \max_{x \sim x'} \|q(x) - q(x')\|_1 = \sum_{i=1}^n |y_i - y'_i|$$

- **Theorem:** Let $q: \mathcal{X}^n \rightarrow \mathbb{R}^d$. The mechanism $\mathbf{M}(x) = q(x) + (z_1, \dots, z_d)$ where each $z_i \leftarrow L(0, \Delta_1/\varepsilon)$ satisfies ε -DP



Approximate Differential Privacy



Definition: Mechanism $\mathbf{M}$: $\mathcal{X}^n \times \mathcal{Q} \rightarrow \mathcal{Y}$ gives (ϵ, δ) -**differential privacy** if for all pairs of **adjacent** datasets $x, x' \in \mathcal{X}^n$, and for all queries $q \in \mathcal{Q}$:

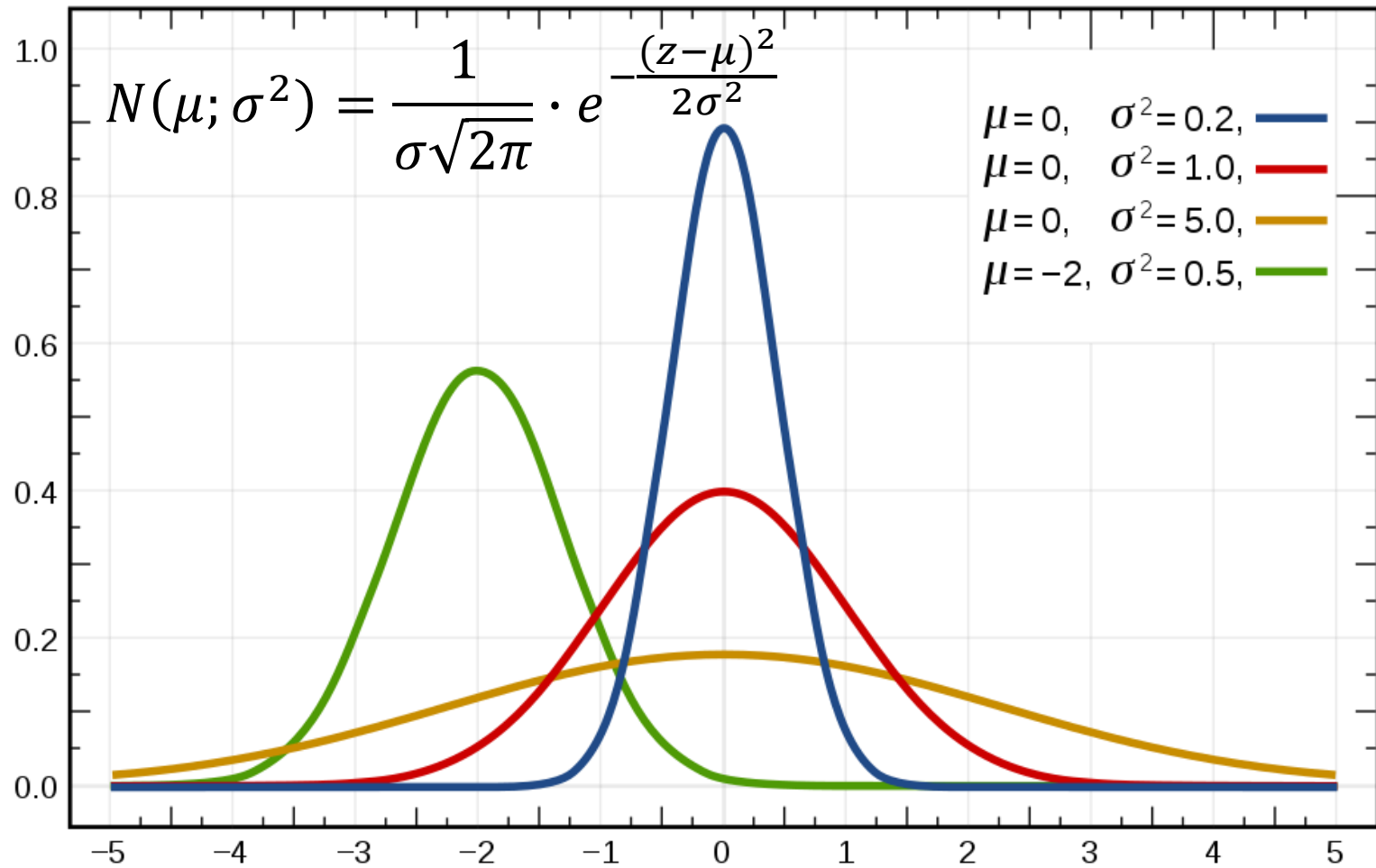
$$\forall y \in \mathcal{Y}, \mathbb{P}[\mathbf{M}(x, q) = y] \leq e^\epsilon \cdot \mathbb{P}[\mathbf{M}(x', q) = y] + \delta$$


Benefits of the Relaxation

- **Gaussian** noise
 - Leading to **better accuracy**
- **Advanced composition**
 - Can answer k queries with **cumulative** loss $\sqrt{k} \cdot \epsilon$
 - Instead of $k\epsilon$ as in **pure** differential privacy
- Can use cryptography to **simulate trusted center**



Gaussian Distribution



Gaussian Mechanism

- Let $q: \mathcal{X}^n \rightarrow \mathbb{R}^d$. The **global ℓ_2 -sensitivity** is:

$$\Delta_2 = \max_{x \sim x'} \|q(x) - q(x')\|_2 = \sum_{i=1}^n \sqrt{(y_i - y'_i)^2}$$

- Theorem:** Let $q: \mathcal{X}^n \rightarrow \mathbb{R}^d$. The mechanism $\mathbf{M}(x) = q(x) + (z_1, \dots, z_d)$ where **each value** $z_i \sim N\left(0, \frac{2 \ln(1.25/\delta) \cdot \Delta_2^2}{\varepsilon^2}\right)$ satisfies (ε, δ) -DP



Gaussian versus Laplace

- Note that for **every** vector $y \in \mathbb{R}^d$, $\|y\|_2 \leq \|y\|_1 \leq \sqrt{d} \cdot \|y\|_2$
- Suppose that $x \in \{0,1\}^{n \times d}$ and take the query $q(x) = \frac{1}{n} \cdot \sum_{i=1}^n x_i$ for the **multivariate mean**
 - Here, $\Delta_1 \leq d/n$ and $\Delta_2 \leq \sqrt{d}/n$
 - The Laplace mechanism would add noise of magnitude $O(d/n\varepsilon)$ whereas the Gaussian mechanism needs **less noise** $O(\sqrt{d}/n\varepsilon)$ for roughly the **same accuracy**



Gaussian Mechanism: Privacy

- We first show that for $\mathbf{M}(x) = q(x) + z$ where $z \leftarrow N(0, \sigma^2 \cdot I)$ the **privacy loss** is **distributed as**

$$N\left(\frac{\|q(x) - q(x')\|_2^2}{2\sigma^2}, \frac{\|q(x) - q(x')\|_2^2}{\sigma^2}\right)$$

$$\begin{aligned} \ln\left(\frac{\mathbb{P}[\mathbf{M}(x) = q(x) + z]}{\mathbb{P}[\mathbf{M}(x') = q(x) + z]}\right) &= \ln\left(\frac{\exp(-\|z\|_2^2/2\sigma^2)}{\exp(-\|z + v\|_2^2/2\sigma^2)}\right) \\ &= -\frac{1}{2\sigma^2} \cdot (\|z\|_2^2 - \|z + v\|_2^2) \\ &= -\frac{1}{2\sigma^2} \cdot \left(\sum_{j=1}^d (z_j^2 - (z_j + v_j)^2)\right) \end{aligned}$$



Gaussian Mechanism: Privacy

- **Fact:** $a \cdot N(0,1) + b \cdot N(0,1) \sim N(0, a^2 + b^2)$
- Simplifying, we get:

$$\ln \left(\frac{\mathbb{P}[\mathbf{M}(x) = q(x) + z]}{\mathbb{P}[\mathbf{M}(x') = q(x) + z]} \right) = \frac{1}{2\sigma^2} \cdot \left(\sum_{j=1}^d (2z_j v_j + v_j^2) \right)$$

- The **constant** term is $\frac{\|v\|_2^2}{2\sigma^2}$ and matches the **mean**
- The **other** term is $\frac{1}{\sigma^2} \cdot \sum_j z_j v_j = \sum_j z'_j = z'$, where $z'_j \sim N(0, \sigma^2 \cdot v_j^2)$ and $z' \sim N\left(0, \frac{\|v\|_2^2}{\sigma^2}\right)$



Gaussian Mechanism: Privacy

- To finish the proof, we need to show that the **privacy loss** is $\leq \varepsilon$ with **probability** $\geq 1 - \delta$
- For $\tilde{z} \sim N(0,1)$ and $\sigma = \Delta_2 \cdot t/\varepsilon$, we can write:

$$\begin{aligned} & \mathbb{P} \left[\left| \frac{\|q(x) - q(x')\|_2}{2\sigma} \cdot \tilde{z} + \frac{\|q(x) - q(x')\|_2^2}{2\sigma^2} \right| \geq \varepsilon \right] \\ &= \mathbb{P} \left[|\tilde{z}| \geq \frac{\varepsilon\sigma}{\|q(x) - q(x')\|_2} - \frac{\|q(x) - q(x')\|_2}{2\sigma} \right] \\ &\leq \mathbb{P} \left[|\tilde{z}| \geq t - \frac{\varepsilon}{2t} \right] \end{aligned}$$

- For simplicity we **drop** the **small term** $\varepsilon/(2t)$



Gaussian Mechanism: Privacy

- By standard **tail bounds** $\mathbb{P}[|\tilde{z}| \geq t] \leq e^{-t^2/2}$
- If we **set** $t = \sqrt{2\ln(2/\delta)}$ we obtain that $\mathbb{P}[|\tilde{z}| \geq t] \leq \delta$, which implies (ε, δ) -DP
- Note that the latter corresponds **roughly** to

$$\sigma \approx \frac{\Delta_2}{\varepsilon} \cdot \sqrt{\ln(1/\delta)}$$



Properties of Approximate DP

- **Post processing**: If $\mathbf{M}: \mathcal{X}^n \times \mathcal{Q} \rightarrow \mathcal{Y}$ is (ε, δ) -DP, and $F: \mathcal{Y} \rightarrow \mathcal{Z}$ is any **randomized function**, then $F \circ \mathbf{M}: \mathcal{X}^n \times \mathcal{Q} \rightarrow \mathcal{Z}$ is (ε, δ) -DP
- **Group privacy**: If $\mathbf{M}$ is (ε, δ) -DP, then for all pairs of datasets $x, x' \in \mathcal{X}^n$, $\mathbf{M}(x)$ and $\mathbf{M}(x')$ are $(k\varepsilon, k\delta \cdot e^{(k-1)\varepsilon})$ -DP for $k = d(x, x')$
- **Basic composition**: If $\mathbf{M}_1, \dots, \mathbf{M}_k$ are (ε, δ) -DP, then $\mathbf{M}$ s.t. $\mathbf{M}(x) = (\mathbf{M}_1(x), \dots, \mathbf{M}_k(x))$ is $(k\varepsilon, k\delta)$ -DP



Advanced Composition

- **Theorem:** For all $\varepsilon, \delta, \delta' > 0$, if $\mathbf{M}_1, \dots, \mathbf{M}_k$ are (ε, δ) -DP, then $\mathbf{M}(x) = (\mathbf{M}_1(x), \dots, \mathbf{M}_k(x))$ is $(\tilde{\varepsilon}, \tilde{\delta})$ -DP for

$$\begin{aligned}\tilde{\varepsilon} &= \varepsilon \sqrt{2k \cdot \log(1/\delta')} + k\varepsilon \cdot \frac{e^\varepsilon - 1}{e^\varepsilon + 1} \\ \tilde{\delta} &= k\delta + \delta'\end{aligned}$$

- In the **high-privacy** regime, $(e^\varepsilon - 1)/(e^\varepsilon + 1) \approx \varepsilon/2$ and thus we can **ignore** the **second term** in $\tilde{\varepsilon}$
- The above holds even if in the **adaptive** setting



Reduction to Binary(ish) Mechanisms

- What is the **simplest** pair of random variables (U, V) satisfying the definition of (ϵ, δ) -DP?
 - I.e., with probability $\geq 1 - \delta$

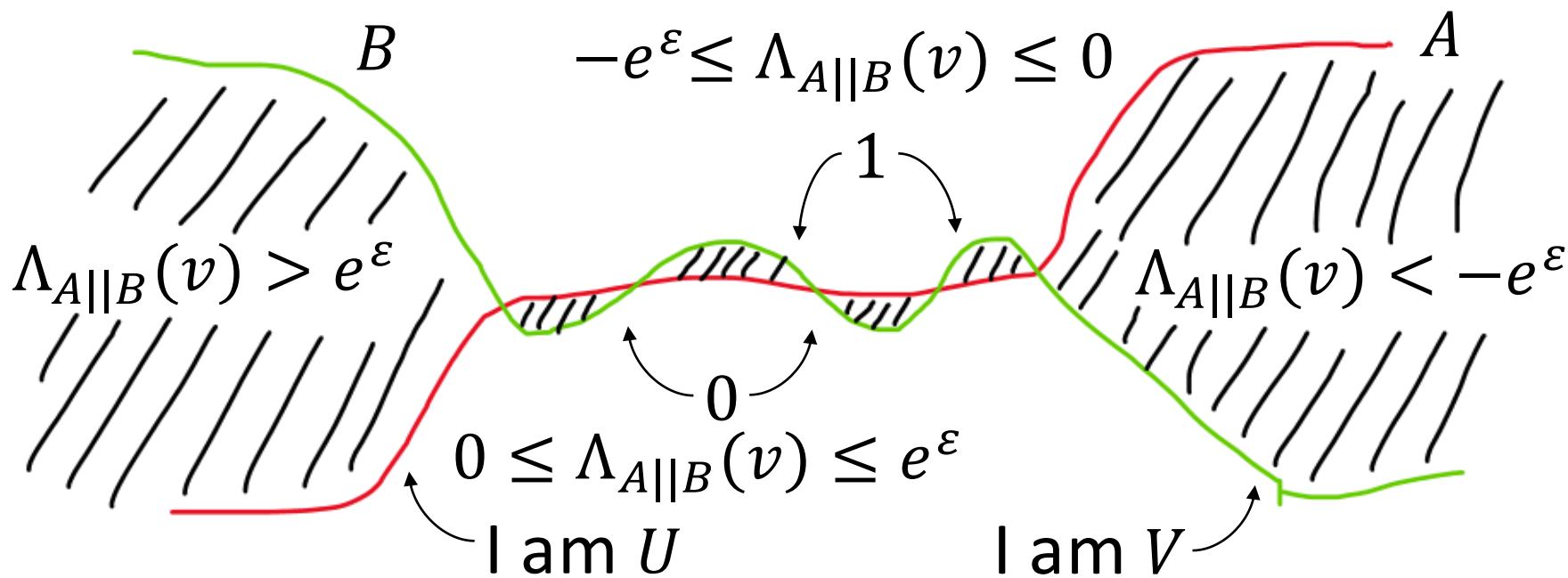
$$\Lambda_{U||V} = \left| \ln \left(\frac{\mathbb{P}[U = v]}{\mathbb{P}[V = v]} \right) \right| \leq \epsilon$$

v	$\mathbb{P}[U = v]$	$\mathbb{P}[V = v]$
0	$e^\epsilon(1 - \delta)/(1 + e^\epsilon)$	$(1 - \delta)/(1 + e^\epsilon)$
1	$(1 - \delta)/(1 + e^\epsilon)$	$e^\epsilon(1 - \delta)/(1 + e^\epsilon)$
I am U	δ	0
I am V	0	δ



Reduction to Binary(ish) Mechanisms

- **Lemma:** Let A and B be s.t. $|\Lambda_{A||B}| \leq \varepsilon$ and $|\Lambda_{B||A}| \leq \varepsilon$ w.p. $1 - \delta$; there is a **randomized mapping** Ψ s.t. $\Psi(A) \sim U$ and $\Psi(B) \sim V$



Reduction to Binary(ish) Mechanisms

- We can think $\mathbf{M}_j$ takes as input $x \in \mathcal{X}^n$ as well as the **transcript** τ_{j-1} of **previous outputs**
- **Corollary:** There is a **randomized mapping** Ψ^* s.t. $\mathbf{M}(x) = (\mathbf{M}_1(x), \dots, \mathbf{M}_k(x))$ satisfies
 - $\mathbf{M}(x) \sim \Psi^*(U_1, \dots, U_k)$, with $U_1, \dots, U_k \sim U$
 - $\mathbf{M}(x') \sim \Psi^*(V_1, \dots, V_k)$, with $V_1, \dots, V_k \sim V$
- By **post-processing**, it suffices to bound the privacy loss between $U_1, \dots, U_k$ and $V_1, \dots, V_k$



Composition of Binary(ish) Mechanisms

- Let $v_j \in \{0,1, \text{I am } U\}$ be the j -th **realization**
 - That is, $v_j \sim U$
 - When $v_j = \text{I am } U$, **privacy is violated**, but
Call this event E_1
 $\mathbb{P}[\exists v_j \text{ s.t. } v_j = \text{I am } U] = 1 - (1 - \delta)^k \leq k\delta$
 - Next, we condition on E_1 **not happening**

$$\ln \left(\frac{\mathbb{P}[(U_1, \dots, U_k) = v]}{\mathbb{P}[(V_1, \dots, V_k) = v]} \right) = \sum_{j=1}^k \ln \left(\frac{\mathbb{P}[U_j = v_j]}{\mathbb{P}[V_j = v_j]} \right)$$
$$\sum_{j=1}^k \ln \left(\frac{(1 - \delta)e^{\varepsilon(1-v_j)} / (e^\varepsilon + 1)}{(1 - \delta)e^{\varepsilon v_j} / (e^\varepsilon + 1)} \right) = \sum_{j=1}^k \varepsilon(1 - 2v_j)$$



Composition of Binary(ish) Mechanisms

- Note that (always conditioning on $\overline{E_1}$)

$$1 - 2v_j = \begin{cases} 1 & \text{w. p. } e^\varepsilon / (1 + e^\varepsilon) \\ -1 & \text{w. p. } 1 / (1 + e^\varepsilon) \end{cases}$$

- Hence, we can compute the **expectation**

$$\mathbb{E} \left[\ln \left(\frac{\mathbb{P}[(U_1, \dots, U_k) = v]}{\mathbb{P}[(V_1, \dots, V_k) = v]} \right) \right] = k\varepsilon \cdot \frac{e^\varepsilon - 1}{e^\varepsilon + 1}$$

- Finally, we apply the **Chernoff bound** in order to prove that the **privacy loss** does **not exceed** its **expectation** with probability more than δ'



Composition of Binary(ish) Mechanisms

- **Hoeffding bound:** For $X_1, \dots, X_k$ i.i.d. and **bounded** in the range $[a, b]$, we have:

$$\mathbb{P} \left[\sum_{j=1}^k X_j \geq \mathbb{E}[X_j] + \gamma \right] \leq e^{-\frac{2\gamma^2}{k(b-a)^2}}$$

- Define the event that the privacy loss **goes too far** from its **mean**

Call this event E_2

$$\ln \left(\frac{\mathbb{P}[(U_1, \dots, U_k) = v]}{\mathbb{P}[(V_1, \dots, V_k) = v]} \right) > k\varepsilon \cdot \frac{e^\varepsilon - 1}{e^\varepsilon + 1} + \beta\varepsilon\sqrt{k}$$



Composition of Binary(ish) Mechanisms

- By **setting** $[a, b] = [-\varepsilon, \varepsilon]$ and $\gamma = \beta\varepsilon\sqrt{k}$

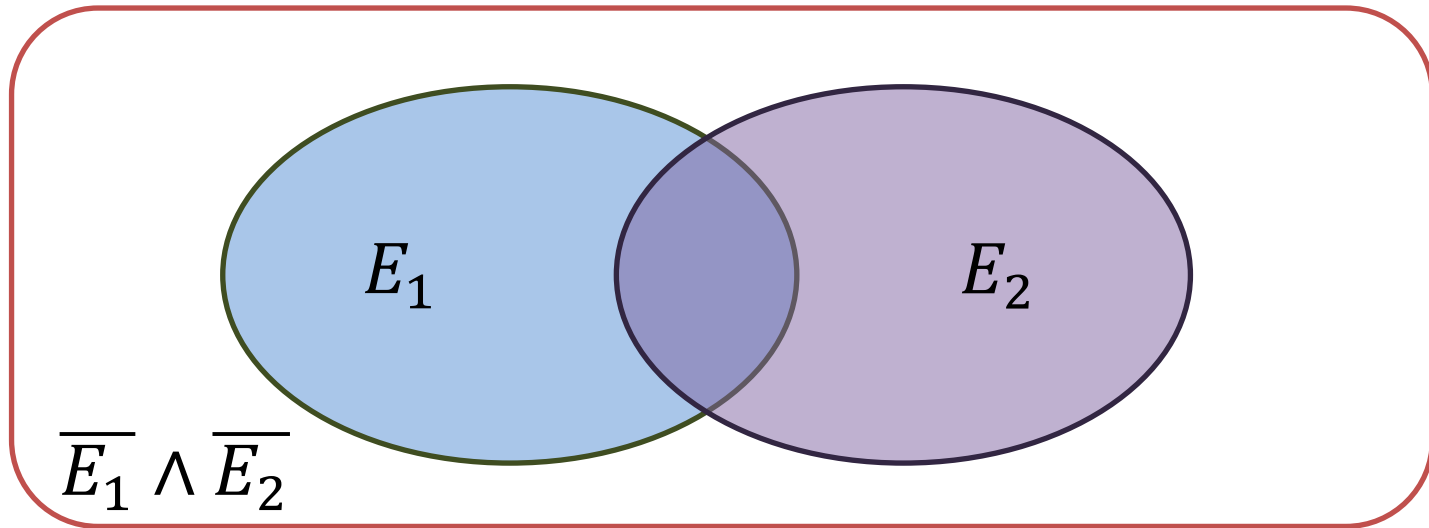
$$\mathbb{P}[E_2 | \overline{E_1}] \leq e^{-\beta^2/2}$$

- Putting it **all together** using $\beta = \sqrt{2\ln(1/\delta')}$

$$\begin{aligned} & \mathbb{P}[(U_1, \dots, U_k) = v \wedge \overline{E_1} \wedge \overline{E_2}] \\ & \leq e^{\tilde{\varepsilon}} \cdot \mathbb{P}[(V_1, \dots, V_k) = v \wedge \overline{E_1} \wedge \overline{E_2}] \\ & \leq e^{\tilde{\varepsilon}} \cdot \mathbb{P}[(V_1, \dots, V_k) = v] \end{aligned}$$



Composition of Binary(ish) Mechanisms



$$\begin{aligned}\mathbb{P}[U^* = v] &= \mathbb{P}[U^* = v \wedge \overline{E_1} \wedge \overline{E_2}] + \mathbb{P}[U^* = v \wedge E_1] \\ &+ \mathbb{P}[U^* = v \wedge \overline{E_1} \wedge E_2] \leq \mathbb{P}[U^* = v \wedge \overline{E_1} \wedge \overline{E_2}] + \mathbb{P}[E_1] \\ &+ \mathbb{P}[E_2 | \overline{E_1}] \cdot \mathbb{P}[\overline{E_1}] \\ &\leq e^{\tilde{\epsilon}} \cdot \mathbb{P}[V^* = v] + k\delta + e^{-\beta^2/2} \cdot 1 \\ e^{\tilde{\epsilon}} \cdot \mathbb{P}[V^* = v] + k\delta + \delta' &= e^{\tilde{\epsilon}} \cdot \mathbb{P}[V^* = v] + \tilde{\delta}\end{aligned}$$



Exponential Mechanism

- Until now, we focused on **numerical** queries
- In some situations, we wish to output **objects**
- Example: **Digital auction**
 - One seller having **infinite copies** of digital good
 - n buyers each with **valuation** v_i
 - What's the price p max. **the revenue** $\sum_{i:v_i \leq p} p$?
- Idea: Use **differential privacy**
 - If $v_1 = v_2 = 1$ and $v_3 = 3.01$, the **revenue drops** from 3 to 1.01 **increasing** p from 1 to 1.01



Exponential Mechanism

- More formally, the mechanism takes as input
 - A **dataset** $x \in \mathcal{X}^n$, a set of **objects** $\mathcal{H}$ and a **score function** $s: \mathcal{X}^n \times \mathcal{H} \rightarrow \mathbb{R}$
 - Only the **dataset is private**
- Define the **sensitivity** of the score function:

$$\Delta s = \max_{h \in \mathcal{H}} \max_{x, x': x \sim x'} |s(x, h) - s(x', h)|$$

- **Definition:** The **exponential mechanism** outputs $h \in \mathcal{H}$ w.p. $\propto \exp(\varepsilon \cdot s(x, h)/(2\Delta s))$



Exponential Mechanism: Privacy

- **Theorem:** The **exponential mechanism** is ε -differentially private
- Fix any $x \sim x'$ and $h \in \mathcal{H}$

$$\begin{aligned} \frac{\mathbb{P}[\mathbf{M}(x, \mathcal{H}, s) = y]}{\mathbb{P}[\mathbf{M}(x', \mathcal{H}, s) = y]} &= \frac{\frac{\exp(\varepsilon \cdot s(x, h)/(2\Delta s))}{\sum_{h' \in \mathcal{H}} \exp(\varepsilon \cdot s(x, h')/(2\Delta s))}}{\frac{\exp(\varepsilon \cdot s(x', h)/(2\Delta s))}{\sum_{h' \in \mathcal{H}} \exp(\varepsilon \cdot s(x', h')/(2\Delta s))}} \\ &= \exp(\varepsilon \cdot (s(x, h) - s(x', h))/(2\Delta s)) \cdot \frac{\sum_{h' \in \mathcal{H}} \exp(\varepsilon \cdot s(x', h')/(2\Delta s))}{\sum_{h' \in \mathcal{H}} \exp(\varepsilon \cdot s(x, h')/(2\Delta s))} \\ &\leq \exp(\varepsilon/2) \cdot \frac{\sum_{h' \in \mathcal{H}} \exp(\varepsilon/2) \cdot \exp(\varepsilon \cdot s(x, h')/(2\Delta s))}{\sum_{h' \in \mathcal{H}} \exp(\varepsilon \cdot s(x, h')/(2\Delta s))} = \exp(\varepsilon) \end{aligned}$$



Exponential Mechanism: Accuracy

- **Theorem**: Let $s^*(x) = \max_{h \in \mathcal{H}} s(x, h)$ and $\mathcal{H}^*$ be the set containing all $h \in \mathcal{H}$ such that $s(x, h) = s^*(x)$. Then:

$$\mathbb{P} \left[s(\mathbf{M}(x, \mathcal{H}, s)) \leq s^*(x) - \frac{2\Delta s}{\varepsilon} \cdot \left(\ln \left(\frac{|\mathcal{H}|}{|\mathcal{H}^*|} \right) + \beta \right) \right] \leq e^{-\beta}$$

- **Corollary**: Since $|\mathcal{H}^*| \geq 1$, we get

$$\mathbb{P} \left[s(\mathbf{M}(x, \mathcal{H}, s)) \leq s^*(x) - \frac{2\Delta s}{\varepsilon} \cdot (\ln(|\mathcal{H}|) + \beta) \right] \leq e^{-\beta}$$



Exponential Mechanism: Accuracy

- By **definition**:

$$\begin{aligned} & \mathbb{P}[s(\mathbf{M}(x, \mathcal{H}, s)) \leq \gamma] \\ &= \frac{\sum_{h \in \mathcal{H}: s(x, h) \leq \gamma} \exp(\varepsilon \cdot s(x, h)/(2\Delta s))}{\sum_{h' \in \mathcal{H}} \exp(\varepsilon \cdot s(x, h')/(2\Delta s))} \\ &= \frac{|\mathcal{H}| \cdot \exp(\varepsilon \gamma/(2\Delta s))}{|\mathcal{H}^*| \cdot \exp(\varepsilon s^*(x, h)/(2\Delta s))} \\ &\leq \frac{|\mathcal{H}|}{|\mathcal{H}^*|} \cdot \exp(\varepsilon(\gamma - s^*(x, h))/(2\Delta s)) \end{aligned}$$

– Set $\gamma = s^*(x) - 2\Delta s/\varepsilon \cdot (\ln(|\mathcal{H}|/|\mathcal{H}^*|) + \beta)$



Application: Laplace Mechanism

- Let $x \in \mathcal{X}^n$ and $q: \mathcal{X}^n \rightarrow \mathbb{R}$ with **sensitivity** Δ
 - Set x to be the **dataset**, $\mathbb{R} = \mathcal{H}$ be the **objects**, and $s(x, h) = -|q(x) - h|$ be the **score**

$$\mathbb{P}[\mathbf{M}(x, \mathcal{H}, s) = h] \propto \exp(\varepsilon \cdot -|q(x) - h|/(2\Delta s))$$

- The latter is identical to the **Laplace mechanism** up to a factor of 2 (resulting in **twice** the noise)
- The factor of 2 can in fact be **removed** by revisiting the **privacy proof**



Application: Selling Digital Goods

- Example: **Digital auction**
 - One seller having **infinite copies** of digital good
 - n buyers each with **valuation** $v_i \in [0,1]$
 - Price $p \in [0,1]$ maxim. **the revenue** $p \cdot |i: v_i \leq p|$
- We first **discretize** $\mathcal{H} = \{\alpha, 2\alpha, \dots, 1\}$ for some α , so that $|\mathcal{H}| = 1/\alpha$
 - Letting $p^* = \max_p p \cdot |i: v_i \leq p|$, we get $s^*(v_1, \dots, v_n) \geq p^*$
 - αn (**round down** p to the **closest multiple** of α , **loosing at most** αn)



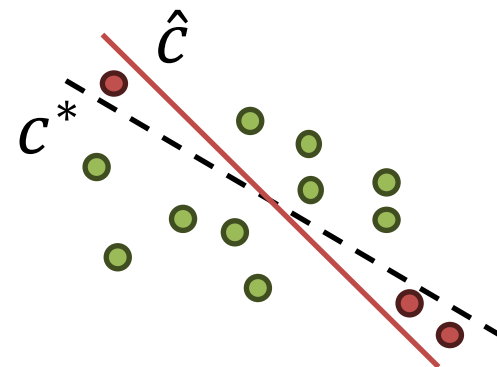
Application: Selling Digital Goods

- Example: **Digital auction**
 - One seller having **infinite copies** of digital good
 - n buyers each with **valuation** $v_i \in [0,1]$
 - Price $p \in [0,1]$ maxim. **the revenue** $p \cdot |i: v_i \leq p|$
- We let $s(x, p)$ be **the revenue** $p \cdot |i: v_i \leq p|$
 - Since $p \leq 1$ and changing **an individual** only affects $|i: v_i \leq p|$ by one, $\Delta s \leq 1$
 - Thus, $s(x, p)$ is **at least** $p^* - \alpha n - \ln(1/\alpha)/\varepsilon$ resulting in $p^* - \ln(n)/\varepsilon$ when $\alpha = \ln(n)/(n\varepsilon)$



Application: Private PAC Learning

- Probably **approximately** correct **learning**
 - **Concept** class $\mathcal{C} = \{c: \{0,1\}^d \rightarrow \{0,1\}\}$
 - n elements $(x_i, y_i = c^*(x_i))$ for some (**unknown**) $c^* \in \mathcal{C}$, where $x_i \sim D$ (also **unknown**)
- **Goal:** Output $\hat{c}$ s.t. $\mathbb{P}_{x \sim D}[\hat{c}(x) \neq c^*(x)]$ is **minimized**
- Example: **Learning halfspaces**
 - Classic task in **machine learning**
 - Intractable **with noise**



Application: Private PAC Learning

- **Theorem:** If $n = \Omega(\log(|\mathcal{C}|)/2\alpha^2)$, then there exists $\hat{c}$ s.t. $\mathbb{P}_{x \sim D}[\hat{c}(x) \neq c^*(x)] \leq \alpha/2$
- Get the **training data** and see how **every function** in $\mathcal{C}$ **classifies** the dataset
 - Output any function $\hat{c} \in \mathcal{C}$ that **never errs**
- Fix $h \in \mathcal{C}$, $n = 2t/\alpha^2$. By a **Chernoff bound**:

$$\mathbb{P}_{x_1, \dots, x_n \sim D} \left[\left| \mathbb{P}_{x \sim D}[h(x) = c^*(x)] - \frac{|i: h(x_i) = c^*(x_i)|}{n} \right| \geq \frac{\alpha}{2} \right] \leq e^{-t}$$

- With $t = \Omega(\log(|\mathcal{C}|))$, the above holds $\forall h \in \mathcal{C}$



Application: Private PAC Learning

- Turning to **differential privacy**, for $x \sim x'$ we have the change of a **single** point (x'_i, y'_i)
 - **Worst case**: x'_i may **not** follow D and $y'_i \neq c^*(x'_i)$
- **Theorem**: If $n = \Omega(2\log(|\mathcal{C}|)/\alpha^2 + \log(|\mathcal{C}|)/(4\alpha\varepsilon))$, then $\exists$ an ε -**DP** algorithm outputting $\hat{c}$ s.t. $\mathbb{P}_{x \sim D}[\hat{c}(x) \neq c^*(x)] \leq \alpha$
- Apply the **exponential mechanism**
 - Set $\mathcal{C} = \mathcal{H}$ and $s((x, y), h) = -|i: h(x_i) \neq y_i|/n$
 - So, $\Delta s = 1/n$ and $s^*((x, y)) = 0$



Application: Private PAC Learning

- The **exponential mechanism** guarantees that we output $\hat{c}$ such that **w.h.p.**:

$$s((x, y), \hat{c}) \geq -\frac{2\Delta s}{\varepsilon} \cdot \ln(|\mathcal{H}|) = -\frac{2}{\varepsilon n} \cdot \ln(|\mathcal{C}|) \geq -\alpha/2$$

- Thus $|i: h(x_i) \neq y_i|/n \leq \alpha/2$
- Putting the two together: $\mathbb{P}_{x \sim D}[\hat{c}(x) \neq c^*(x)] \leq \alpha/2 + \alpha/2 = \alpha$



Answering Many Queries

- Assume we are given a set Q of $k = |Q|$ queries, and we wish to answer all with ε -DP
 - First add Laplace noise to achieve ε_0 -DP
 - By **basic composition** can set $\varepsilon_0 = \varepsilon/k$ so that the noise per query has scale $O(1/(\varepsilon_0 n)) = O(k/\varepsilon n)$
- The above implies that we can answer **all queries** in Q with ε -DP to within error

$$\alpha \leq O\left(\frac{k \log k}{\varepsilon n}\right)$$

Setting $\beta = 1/O(k)$



Answering Many Queries

- For (ε, δ) -DP, we can use the **Gaussian** mechanism and **advanced composition** to get:

$$\alpha \leq O\left(\frac{\sqrt{k \log k \cdot \log(1/\delta)}}{\varepsilon n}\right)$$

- Thus, we can **accurately** answer $k = o(n)$
- However, note that whenever $|Q|$ is **larger** than n^2 , the error is **too large**
- Next, we show how to answer **much more** than n^2 **counting queries**



SmallDB Mechanism

- **Theorem:** There exists an ε -DP mechanism $\mathbf{M}$ such that for all datasets $x \in \mathcal{X}^n$ w.h.p. $\mathbf{M}(x)$ answers **all queries** in $\mathcal{Q}$ to within error

$$\alpha \leq O\left(\frac{\log |\mathcal{X}| \cdot \log |\mathcal{Q}|}{\varepsilon n}\right)^{1/3}$$

Can handle $\gg n^2$
queries

- Moreover, $\mathbf{M}(x)$ outputs a **synthetic dataset** $y \in \mathcal{X}^m$ with $m = O(\log |\mathcal{Q}|/\alpha^2)$ s.t. $\forall q \in \mathcal{Q}$ w.h.p. $|q(y)| \leq \alpha$



SmallDB Mechanism

- **For each** $y \in \mathcal{X}^m$, let
$$\text{weight}_x(y) = \exp(-\varepsilon n \cdot \max_{q \in \mathcal{Q}} |q(y) - q(x)|)$$
- Output y w.p. $\propto \text{weight}_x(y)$, i.e.

$$\Pr[\mathbf{M}(x) = y] = \frac{\text{weight}_x(y)}{\sum_{z \in \mathcal{X}^m} \text{weight}_x(z)}$$

- **Exponential mechanism** with **dataset** $x \in \mathcal{X}^n$, **objects** $\mathcal{H} = \{y = O(\log |\mathcal{Q}|/\alpha^2)\}$, and **score function** $s(x, y) = -\max_{q \in \mathcal{Q}} |q(y) - q(x)|$



SmallDB Mechanism: Privacy & Accuracy

- **Corollary:** The SmallDB mechanism is 2ε -DP
 - The proof follows directly from the **privacy property** of the **exponential mechanism**
- The **accuracy** proof is more involved
 - First, we show there is at least one **good small dataset** $y \in \mathcal{X}^m$ s.t. $|s(x, y)| \leq \alpha$
 - Then, we show the **exponential mechanism** outputs such a **good** dataset **w.h.p.**



SmallDB Mechanism: Accuracy

- **Chernoff bound**: For $X_1, \dots, X_m$ i.i.d. in $[0,1]$ and $X = \sum_{j=1}^m X_j$ with $\mu = \mathbb{E}[X]$

$$\mathbb{P}[X \geq \mu + \varepsilon] \leq e^{-2m\varepsilon^2} \text{ and } \mathbb{P}[X \leq \mu - \varepsilon] \leq e^{-2m\varepsilon^2}$$

- Let y^* be a **random sample** of m rows from x
 - Then $q(y^*) = \sum_{j=1}^m q(x_j)$ and $\mathbb{E}[q(y^*)] = q(x)$
- By the **union bound**, and invoking the Chernoff bound with $m = O(\log |Q|/\alpha^2)$:

$$\Pr[\exists q \in Q \text{ s.t. } |q(y^*) - q(x)| > \alpha] \leq 2|Q| \cdot 2^{-2m\alpha^2}$$



SmallDB Mechanism: Accuracy

- By **accuracy** of the **exponential mechanism** with $\Delta s = 1/n$ and $|\mathcal{H}| = |\mathcal{X}|^{\log |\mathcal{Q}|/\alpha^2}$

$$\mathbb{P} \left[s(\mathbf{M}(x, \mathcal{H}, s)) \leq s^*(x) - \frac{2}{\varepsilon n} \cdot \left(\frac{\log |\mathcal{X}| \cdot \log |\mathcal{Q}|}{\alpha^2} + \log(1/\beta) \right) \right] \leq \beta$$

$$\Rightarrow \mathbb{P} \left[\max_{q \in \mathcal{Q}} |q(y) - q(x)| \geq \alpha + \frac{2}{\varepsilon n} \cdot \left(\frac{\log |\mathcal{X}| \cdot \log |\mathcal{Q}|}{\alpha^2} + \log(1/\beta) \right) \right] \leq \beta$$



SmallDB Mechanism: Accuracy

- By replacing α with $\alpha/2$ and **setting** $\alpha/2 = 2/(\varepsilon n)$
• $\left(\frac{4 \log |\mathcal{X}| \cdot \log |\mathcal{Q}|}{\alpha^2} + \log(1/\beta) \right)$

$$\mathbb{P} \left[\max_{q \in \mathcal{Q}} |q(y) - q(x)| \geq \frac{\alpha}{2} + \frac{\alpha}{2} = \alpha \right] \leq \beta$$

- Thus, w.p. **at least** $1 - \beta$ the **accuracy** is:

$$\alpha \leq O \left(\frac{\log |\mathcal{X}| \cdot \log |\mathcal{Q}|}{\varepsilon n} \right)^{1/3}$$



The Downside

- The exponential mechanism can be **very expensive**
 - Need to **enumerate over all** $y \in \mathcal{Y}$
- Computation time is

$$\begin{aligned}\Omega(|\mathcal{Y}|) &= \Omega(|\mathcal{X}|^m) \\ &= \Omega(|\mathcal{X}|^{O(\log |\mathcal{Q}|/\alpha^2)})\end{aligned}$$

- Answering all queries in the family $\mathcal{Q}_{\text{conj}}(d)$ **with error tending to zero** requires $n = \omega(d^2/\varepsilon)$



Private Multiplicative Weights

- We now present the state-of-the-art mechanism for **linear queries**
 - Query $q: \mathcal{X} \rightarrow [0,1]$ instead of $q: \mathcal{X} \rightarrow \{0,1\}$
 - For a **dataset** $x \in \mathcal{X}^n$, $q(x) = \frac{1}{n} \sum_{i=1}^n q(x_i)$
- **Theorem:** There is a mechanism that answers a **set** Q of **linear queries** on a dataset with (ϵ, δ) -**DP** and **accuracy**

Running time $\tilde{O}(|Q| \cdot |\mathcal{X}| \cdot n/\alpha^2)$

$$\alpha \leq O\left(\frac{\sqrt{\log |\mathcal{X}| \cdot \log 1/\delta \cdot \log |Q|}}{\epsilon n}\right)^{1/2}$$



Lower Bounds

- So far, we have seen DP mechanisms able to answer **many queries** with **good accuracy**
- Next, we look at **lower bounds** essentially telling that these algorithms are **optimal**
- We will consider both
 - Information-theoretic lower bounds
 - Computational lower bounds



Blatant Non-Privacy

- A mechanism $\mathbf{M}: \mathcal{X}^n \rightarrow \mathcal{Y}$ is **blatantly non-private** if for every $x \in \mathcal{X}^n$, one can use $\mathbf{M}(x)$ to compute $x' \in \mathcal{X}^n$ s.t. x and x' differ in at most $n/10$ coordinates w.h.p.
 - A very **weak privacy notion**, ruling out attacks that can reconstruct **almost all** of the dataset
 - **Remark:** A mechanism that is $(1, .1)$ -DP **cannot be** blatantly non-private



Reconstruction Attacks

- Let $\mathcal{X} = \{0,1\}$, so that a dataset of n people is a vector $x \in \{0,1\}^n$
- Consider normalized **inner-product queries** $q \in \{0,1\}^n$, with answer $\langle q, x \rangle / n \in [0,1]$
 - Bits of x are **attributes** of the n members, and q specifies a subset of the population according to some **demographics**
 - The value $\langle q, x \rangle / n$ measures the **correlation** between the demographics and the attributes
 - Can be **transformed** into **counting queries**



Reconstruction Attacks

- **Theorem**: If we are given **for each** $q \in \{0,1\}^n$ a value $y_q \in \mathbb{R}$ s.t. $|y_q - \langle q, x \rangle / n| \leq \alpha$, then we can use the y_q 's to compute x' **differing** from x in $\leq 4\alpha$ fraction of the coordinates
- **Corollary**: If $\mathbf{M}(x)$ outputs y_q as above with $\alpha \leq 1/40$, then $\mathbf{M}$ is **blatantly non-private**
 - Thus, additive error $\Omega(1)$ is **necessary** for answering all 2^n normalized inner-product queries
 - This shows that the error in SmallDB is **tight**



Reconstruction Attacks

- Pick any x' such that $\forall q: |y_q - \langle q, x' \rangle / n| \leq \alpha$
 - **At least one** x' exists, namely x
- Let $q_1 = x$ and $q_0 = \bar{x}$
- The Hamming distance between x, x' is:

$$\begin{aligned} \frac{d(x, x')}{n} &= \frac{|\langle q_0, x \rangle - \langle q_0, x' \rangle| + |\langle q_1, x \rangle - \langle q_1, x' \rangle|}{n} \\ &\leq \left| \frac{\langle q_0, x \rangle}{n} - y_{q_0} \right| + \left| y_{q_0} - \frac{\langle q_0, x' \rangle}{n} \right| \\ &\quad + \left| \frac{\langle q_1, x \rangle}{n} - y_{q_1} \right| + \left| y_{q_1} - \frac{\langle q_1, x' \rangle}{n} \right| \leq 4 \cdot \alpha \end{aligned}$$



Reconstruction Attacks

- Dinur and Nissim provided a **computationally efficient** variant of the above attack
- **Theorem:** For **every** mechanism that answers **all** normalized inner-product queries with accuracy $O(\alpha\sqrt{n})$, there is an **efficient** attacker that reconstructs the dataset in **all but** $O(\alpha^2)$ positions by asking $O(n)$ queries
 - This shows that the Laplace and Gaussian mechanisms are also **tight**

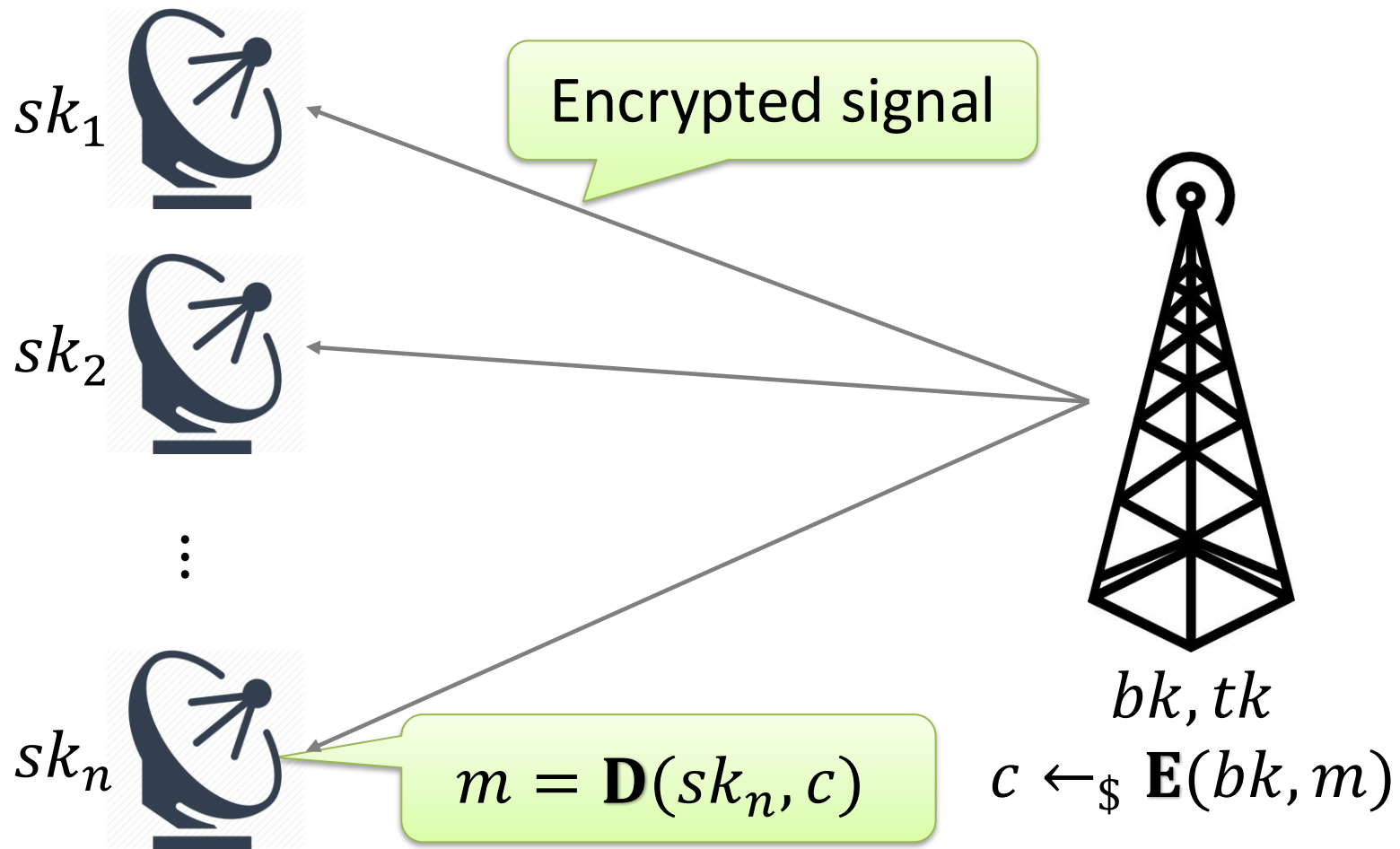


Attacks based on Traitor Tracing

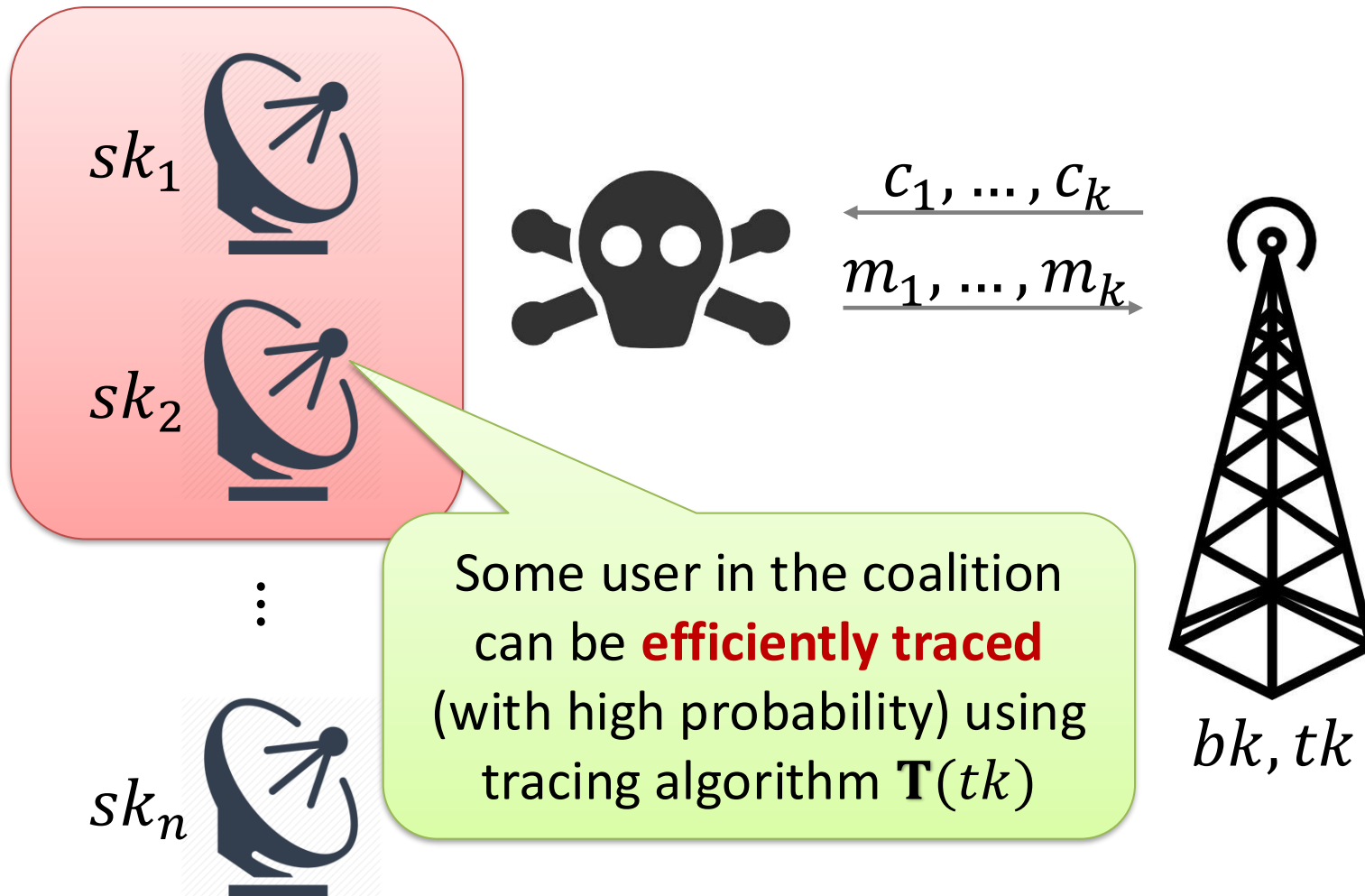
- The smallDB and private multiplicative weight mechanisms answer $\gg n^2$ queries over $\{0,1\}^d$
 - As long as n is large compared to d (e.g., $n \geq d^2$)
- But the computation time is **exponential** in d
- We now show that the above limitation is **inherent** in the **worst case**
- Proof based on **traitor tracing** schemes
 - **Cryptographic** tool for preventing piracy of digital content (using a broadcast channel)



Traitor Tracing



The Tracing Algorithm



Stateless vs Stateful Pirates

- Pirate corrupts any set $S \subseteq [n]$ of decoders and produces a pirate program $\tilde{\mathbf{P}}$
- **Stateless** pirates
 - The pirate program $\tilde{\mathbf{P}}$ is given to the tracer
 - **Useful decryptor:** $\tilde{\mathbf{P}}$ decrypts honest ciphertexts
- **Stateful** pirates
 - The tracing algorithm can query $\tilde{\mathbf{P}}$ on $(c_1, \dots, c_k)$
 - **Cooperativeness:** $\tilde{\mathbf{P}}$ decrypts honest ciphertexts, even after receiving **malformed ciphertexts**



A Computational Lower Bound

- **Theorem:** Assuming **OWFs**, there is a traitor tracing scheme secure against stateful but cooperative pirates
 - Tracing query complexity $k(n, d) = \tilde{O}(n^2)$
- **Theorem:** Every $(1, 1/10n)$ -DP mechanism for answering $k = k(n + 1, d)$ counting queries within error $\alpha < 1/2$ on datasets with n individuals from $\mathcal{X} = \{0,1\}^d$ must run in time **superpolynomial** in d



Proof Sketch

- Let **M** be as in the statement and setup the traitor tracing scheme with $n + 1$ users
- Dataset x contains the **secret keys** $sk_i \in \{0,1\}^d$ of all users but one (chosen at **random**)
- Counting queries: $q_c(sk_i) = \mathbf{D}(sk_i, c)$
 - Hence, **M**(x) yields an $\pm\alpha$ approximation a of the number of users in x whose key decrypts c to 1
 - If c is a **valid encryption** of m , then $|a - m| \leq \alpha < 1/2$ so that **rounding** a equals m



Proof Sketch

- Define the pirate to be

$$\tilde{\mathbf{P}}((sk_i)_{i \in S}, c_1, \dots, c_k) = \lceil \mathbf{M}(x, q_{c_1}, \dots, q_{c_k}) \rceil$$

- The accuracy of $\mathbf{M}$ implies that $\tilde{\mathbf{P}}$ **cooperates**
- Moreover, by **postprocessing**, $\tilde{\mathbf{P}}$ is DP too
- Next, we show that **tracing contradicts DP**
- Thus, $\tilde{\mathbf{P}}$ must **not be traceable** and hence must have **superpolynomial** running time



Proof Sketch

- By **traceability** of the traitor tracing, w.p. ≈ 1 , algorithm $\mathbf{T}^{\tilde{\mathbf{P}}}((sk_i)_{i \in S}, \cdot)(tk)$ outputs $i \in S$
- Thus, for large enough n , there is an i^* s.t.

$$\Pr[\mathbf{T}^{\tilde{\mathbf{P}}}((sk_i)_{i \in S}, \cdot)(tk) = i^*] \geq 1/2n$$

- Let $S' = \{1, \dots, n+1\} \setminus \{i^*\}$; by DP:

$$\begin{aligned} & \Pr[\mathbf{T}^{\tilde{\mathbf{P}}}((sk_i)_{i \in S}, \cdot)(tk) = i^*] \\ & \leq e \cdot \Pr[\mathbf{T}^{\tilde{\mathbf{P}}}((sk_i)_{i \in S'}, \cdot)(tk) = i^*] + 1/10n \end{aligned}$$



Proof Sketch

- Thus,

$$\Pr \left[\mathbf{T}^{\tilde{\mathbf{P}}((sk_i)_{i \in S'}, \cdot)}(tk) = i^* \right] \geq 1/2en - 1/10en \geq \Omega(1/n)$$

- **Corollary:** Assuming **OWFs**, for every $n = \text{poly}(d)$ there is no **poly-time** $(1, 1/10n)$ -DP mechanism for answering more than $\tilde{O}(n^2)$ queries over $\mathcal{X} = \{0,1\}^d$ within $\alpha < 1/2$
 - This is **tight**, as we can **accurately** answer $k = \tilde{\Omega}(n^2)$ counting queries in **polynomial time**



Simple Traitor Tracing

- Let $(\mathbf{E}, \mathbf{D})$ be any **symmetric encryption**
- The broadcast key $bk = (sk_1, \dots, sk_n)$ consists of n **independent secret keys**, and $tk = bk$
- To encrypt $b \in \{0,1\}$, output

$$c = (\mathbf{E}(sk_1, b), \dots, \mathbf{E}(sk_n, b))$$

- To decrypt $c = (c^{(1)}, \dots, c^{(n)})$ use sk_i
 - Suffices to know **which portion** corresponds to the i -th user



How to Trace

- Tracing exploits ciphertexts that **different users** would **decrypt differently**

$$\mathbf{TrE}(sk, i) \\ = (\mathbf{E}(sk_1, 1), \dots, \mathbf{E}(sk_i, 1), \mathbf{E}(sk_{i+1}, 0), \dots, \mathbf{E}(sk_n, 0))$$

- Note that users $j \leq i$ would output 1, but users $j > i$ would output 0



How to Trace

- Consider the matrix below

$$\begin{pmatrix} 00 \dots 00 & 11 \dots 11 & 11 \dots 11 & 11 \dots 11 & 11 \dots 11 \\ & 00 \dots 00 & 11 \dots 11 & 11 \dots 11 & \\ & & 00 \dots 00 & 11 \dots 11 & \\ & & & \dots & \\ 00 \dots 00 & & & 00 \dots 00 & 11 \dots 11 \end{pmatrix}$$

- Encrypt each column and **randomly permute**
 - I.e., generate **random** $i_1, \dots, i_k \in [0, n]$ for $k = (n + 1) \cdot s$ s.t. each of $[0, n]$ appears s times
 - Then $C = (c_j)_{j \in [k]}$ with $c_j \leftarrow_{\$} \mathbf{TrE}(sk, i_j)$



How to Trace

- Next, query $\tilde{\mathbf{P}}((sk_i)_{i \in S}, \cdot)$ with $(c_1, \dots, c_k)$ obtaining $(b_1, \dots, b_k)$, and compute

$$\forall i \in [0, n]: p_i = \frac{1}{s} \cdot \sum_{j: i_j = i} b_j$$

- Output any i^* such that $p_{i^*} - p_{i^*-1} \geq 1/n$
 - Note that if $c \leftarrow_{\$} \mathbf{TrE}(sk, 0)$, then **every user** would return $b = 0$ (similarly for $\mathbf{TrE}(sk, n)$)
 - Thus, $p_0 = 0$ and $p_n = 1$ and so **there exists** i^* such that $p_{i^*} - p_{i^*-1} \geq 1/n$



Analysis

- It remains to show that w.h.p. $i^* \in S$
- Note that $\mathbf{TrE}(sk, i^*)$ and $\mathbf{TrE}(sk, i^* - 1)$ differ for the message encrypted under sk_{i^*}
 - And if $i^* \notin S$ this key is **unknown** to the pirate
- By security of encryption, we can replace k repetitions of $\mathbf{E}(sk_{i^*}, 1)$ with $\mathbf{E}(sk_{i^*}, 0)$ **without effecting** the success of the pirate
 - After this change $\mathbf{TrE}(sk, i^*)$ and $\mathbf{TrE}(sk, i^* - 1)$ are **identical**



Analysis

- Since $i_1, \dots, i_k$ are **random**, the pirate **does not know** which i_j is i^* and which is $i^* - 1$
 - Thus, if it wants to make p_{i^*} larger than p_{i^*-1} , for $i^* \notin S$ it can't do better than **guessing**
- Taking $s = \tilde{O}(n^2)$ and applying Chernoff yields that $\forall i \notin S$ w.h.p. $p_i - p_{i-1} = o(1/n)$
- Note that the query complexity is $k = \tilde{O}(n^3)$
 - The above can be improved to $k = \tilde{O}(n^2)$ by using **fingerprinting codes**



Hardness of Synthetic Data

- Some mechanisms work by producing a **compact representation** of **all answers**
 - This is the case, e.g., of SmallDB
- In the traitor tracing world this corresponds to **stateless pirates**
 - If the ciphertext length is $\ell(n, d)$, the previous proof rules out **efficient** mechanisms for answering families $\mathcal{Q}$ of counting queries of description length $\ell(n + 1, d)$ and size $2^{\ell(n+1, d)}$
 - Interesting only if $\ell \ll n$



Hardness of Synthetic Data

- The above applies only to "**unnatural**" Q
- Towards overcoming this limitation, consider the following database using signature $(\mathbf{S}, \mathbf{V})$
 - Choose **single** sign/verify key pair (vk^*, sk^*)
 - Database made by n rows: $(m_i, \mathbf{S}(sk^*, m_i), vk^*)$ for **random messages** m_i
 - One query for each vk : What fraction of rows are valid signatures w.r.t. vk (i.e., $q_{vk}(\cdot) = \mathbf{V}(vk, \cdot)$)?



Hardness of Synthetic Data

- **Efficient curator** cannot generate **synthetic dataset** which is accurate w.r.t. vk^*
 - Let **M** output $\hat{x} \in (\{0,1\}^d)^{\hat{n}}$
 - By accuracy, $\hat{x}$ contains $\hat{x}_j = (\hat{m}_j, \hat{\sigma}_j)$ such that $\mathbf{V}\left(vk^*, (\hat{m}_j, \hat{\sigma}_j)\right) = 1$
- If $\hat{m}_j \notin x$, then **M** violates unforgeability
- If $\hat{m}_j \in x$, then **M** violates differential privacy
 - For each $i \in [n]$, if **M** has (ϵ, δ) -DP it outputs m_i w.p. $\leq e^\epsilon / 2^d + \delta$



Hardness of Synthetic Data

- Finally, it is possible to express the query $\mathbf{V}(vk^*, \cdot)$ with **2-way conjunctions**
 - By means of the PCP theorem
- **Theorem:** Assuming **OWFs**, there exists $\alpha > 0$ such that there is no $n = \text{poly}(d)$ and **poly-time** $(1, 1/10n)$ -DP mechanism that given dataset $(\{0,1\}^d)^n$ outputs a **synthetic dataset** approximating all the queries in $\mathcal{Q}_{\text{conj}}^2(d)$ to within error at most α



Incentives

- Until now the goal was designing differentially private mechanisms, but the data is assumed to be **already there**
- But why should someone **participate** in the computation?
- Why would they give their **true data**?
- Do we need **compensation**? How much?



Game Theory and Mechanism Design

- Idea: Solve **optimization problem**
- Catch: **No access** to inputs
 - Inputs held by **self-interested** agents
- Design **incentives** and choice of solution (mechanism) that incentivizes **truth-telling**
 - No need for participants to strategize
 - Simple to predict what will happen
 - Often a non-truth-telling mechanism can be replaced by one where the coordinator does the lying on behalf of the participants



Good News

- **Composition**: Approximate truthfulness still satisfied under composition!
- **Collusion resistance**: $O(k\varepsilon)$ -approximate dominant strategy, even for coalitions of k agents
- Both properties not immediate in game-theoretic mechanism design
- All done **without money**!



Bad News

- But not only truthful reporting gives an approximate dominant strategy
 - **Any** report does so, even **malicious** ones
- How do we actually properly get people to **truthfully participate**?
 - Perhaps need **compensation**
 - Much harder to achieve



Differential Privacy as a Tool

- **Nash Equilibrium**: An assignment of players to strategies so that no player would benefit by changing strategy, given how everyone else is playing
- **Correlated Equilibrium**: Players have access to correlated signals (e.g., traffic light)
 - Every Nash equilibrium is a correlated equilibrium, but not viceversa
- Differential privacy has applications to mechanism design with **correlated equilibria**



The Issue of Verification

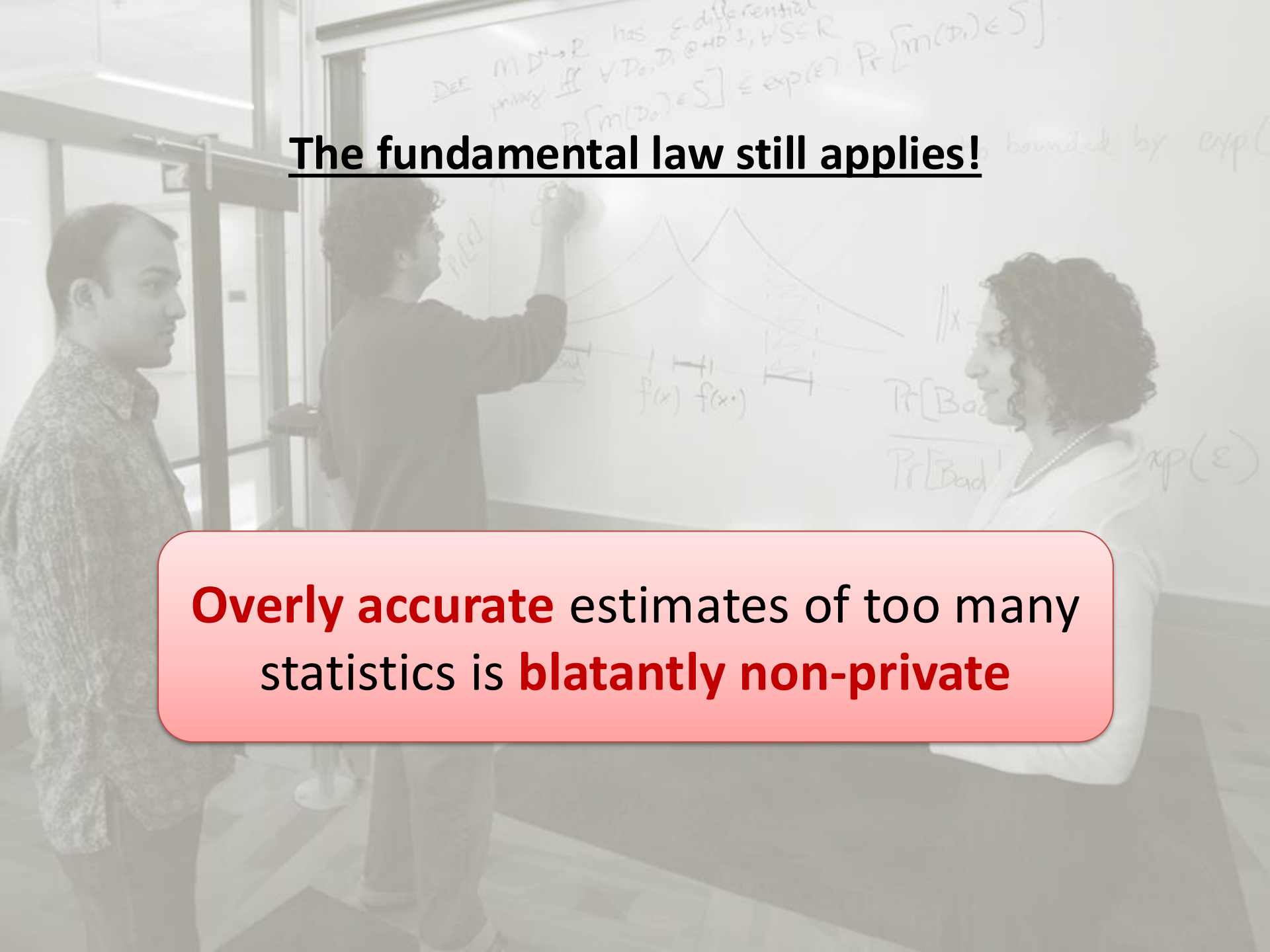
- Challenging to strictly **incentivize truth-telling** in differentially private mechanisms design
- Exceptions:
 - Responses are **verifiable**
 - Agents **care** about outcome
- Challenge: No observed outcome
 - What is the prevalence of drug use?
 - Are students cheating in class?



Privacy and Game Theory

- **Asymptotic truthfulness**, new mechanisms design and equilibrium selection results
- Interesting challenge of modeling **costs for privacy**
- In order to design privacy for humans do we need to understand
 - How people currently value or should value it?
 - What are the right promises to give?





The fundamental law still applies!

Overly accurate estimates of too many statistics is **blatantly non-private**